

BHARATH NALLAMOTHU

Principal AI Engineer

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PROFESSIONAL SUMMARY

Principal AI Engineer with 11+ years of experience designing and delivering production AI and software platforms across automotive, telecommunications, healthcare, and education. Core contributor to Toyota Motor North America Generative AI products and DELight, an AI-powered data intelligence platform spanning multi-source profiling, anomaly detection, AI insights, extraction, and lakehouse workflows. Combines Python, FastAPI, LangGraph, Dagster, RAG, evaluation, AWS, system design, and full-stack delivery to build reliable, scalable, and cost-aware AI systems.

TECHNICAL SKILLS

Agentic AI and Evaluation: LangChain, LangGraph, LangSmith, RAGAS, ReAct, agent and tool orchestration, supervisor and hierarchical agents, context engineering, prompt optimization, human-in-the-loop workflows

Data Intelligence and Retrieval: RAG, OpenSearch, FAISS, HNSW, knowledge graphs, semantic, hybrid and vector search, data profiling, anomaly detection, Text-to-SQL, PostgreSQL, Redis, Databricks

Orchestration and Backend: Python, FastAPI, Dagster, SQLAlchemy, PyMongo, Pandas, CSV, Parquet, execution and run tracking, REST APIs, GraphQL, Node.js

AWS and Engineering: AWS Secrets Manager, Amazon Bedrock, SageMaker, ECS, Lambda, Step Functions, DynamoDB, S3, S3 Vectors, Docker, TypeScript, React, CI/CD

PROFESSIONAL EXPERIENCE

Principal AI Engineer | Toyota Motor North America

Feb 2024 - Present

- Build Python and FastAPI services, LangGraph agent workflows, retrieval pipelines, tool integrations, and production experiences across enterprise Generative AI and data intelligence products; apply evaluation, observability, retries, fallbacks, and human-in-the-loop controls.
- **DELight - orchestration:** Developed FastAPI APIs and Dagster workflows for execution creation and history, multi-source profiling, AI business insights, and anomaly detection across PostgreSQL, MongoDB, file, RDS, and Databricks flows; coordinated parallel AI and anomaly jobs after profiling.
- **DELight - extraction:** Developed backend data-movement workflows using SQLAlchemy and PyMongo cursors, Pandas batching, CSV or Parquet output, and S3; enforced exact output-path validation before downstream structured profiling and, for supported relational flows, reconciliation.
- **DELight - reliability and security:** Implemented execution and run tracking, run-specific polling, bounded 429 retries, terminal-status normalization, duplicate downstream-job protection, and failure guards; used AWS Secrets Manager references and default credential-chain access so browser clients never received stored AWS secrets.

SELECTED TOYOTA GENERATIVE AI PROJECTS

- **GearPal:** Co-developed an agentic RAG troubleshooting assistant with permission-aware OpenSearch routing, hybrid and vector retrieval, metadata filters, API and tool integration, and confidence-based fallback. The team platform supports 20 shops across 7 North American plants and 1,300+ maintenance professionals.
- **ToyotaGPT:** Implemented LangGraph workflows for conditional execution, model and tool selection, structured responses, conversation memory, session restoration, history retrieval, and reusable application integrations.
- **MaterialGPT:** Co-developed multi-agent RAG research across service reports, ANSYS data, technical documentation, and PostgreSQL Text-to-SQL using planning, specialist agents, follow-ups, and citation-backed synthesis. The team platform serves 1,000+ users and reduced initial research discovery from 4-5 days to minutes.
- **BatteryBrainGPT:** Built retrieval and LLM orchestration for a battery engineering assistant connecting technical knowledge sources and APIs for diagnostics, research, and troubleshooting.

Full Stack Developer | Verizon

Oct 2022 - Feb 2024

- Designed TypeScript and Node.js microservices using Express.js, PostgreSQL, REST APIs, GraphQL, Apollo, OAuth 2.0, and NGINX.
- Built React and TypeScript applications with Redux, reusable components, API integrations, code reviews, and Jest and React Testing Library coverage.
- Supported Dockerized AWS deployments, OpenShift, and Jenkins CI/CD for reliable production delivery.

ADDITIONAL EXPERIENCE

Senior Software Developer | Novartis

Oct 2020 - Oct 2022

- Delivered React applications backed by Node.js, GraphQL and Apollo, REST APIs, Jest, AWS, and Docker.
- Built accessible, reusable components and data-rich interfaces, including D3 visualizations and large-data rendering workflows.

Senior Software Engineer - Lead | Study Monitor

May 2018 - Oct 2020

- Led React and React Native delivery using Redux, Node.js and LoopBack, MongoDB, AWS, REST APIs, and automated testing.
- Managed implementation quality and end-to-end mobile release, signing, configuration, and deployment workflows.

Software Developer | LivBIM

Apr 2017 - May 2018

- Built Angular and Node.js and LoopBack applications, reusable components, WebSocket-driven experiences, and AWS-hosted services.
- Delivered responsive interfaces and real-time data workflows across frontend and backend components.

Software Developer | YITSOL

Sep 2015 - Apr 2017

- Developed React, AngularJS, JavaScript, Node.js, and REST-based interfaces.
- Supported Docker, OpenShift, and AWS delivery for partner-facing and back-office applications.

IT Engineer | CMC, A TATA Enterprise

Jun 2014 - Sep 2015

- Built MVC web interfaces using HTML, CSS, JavaScript, jQuery, JSON, and AJAX.
- Implemented dynamic user workflows, client-side validation, and server-side data integration.

EDUCATION

Bachelor's Degree | Jawaharlal Nehru Technological University